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Tropical forest cover density mapping

A. RIKIMARU[1], P.S.ROY[2] & S.MIYATAKE[3]

[1]*Nagaoka University of Technology;* [2]*Indian Institute of Remote Sensing (NRSA), Dehradun, India;* [3]*Japan Overseas Forestry Consultants Association, Tokyo, Japan*

Abstract: Forest canopy density is one of the most useful parameters to consider in the planning and implementation of rehabilitation program. This study is development of bio-physical analysis model for obtaining of Forest Canopy Density (FCD) using LANDSAT TM data image analysis. The components of FCD model are four factors; vegetation, bare soil, thermal and shadow. This work is implemented under the research project; PD32/93Rev2(F) of International Tropical Timber Organization (ITTO).

Resumen: La densidad de dosel forestal es uno de los parámetros cuya inclusión tiene más utilidad en la planeación e instrumentación de programas de rehabilitación. Este estudio consiste en el desarrollo de un modelo de análisis biofísico para la obtención de Densidad de Dosel Forestal (FCD por sus siglas en inglés) por medio del análisis de datos de imágenes LANDSAT TM. Los componentes del modelo FCD son cuatro factores; vegetación, suelo desnudo, temperatura y sombra. Este trabajo forma parte del proyecto de investigación PD32/93Rev2(F) de la Organización Internacional de Maderas Tropicales (ITTO).

Resumo: A densidade do copado florestal é um dos parâmetros mais úteis a serem considerados no planeamento e implementação de programas de reabilitação. Este estudo apresenta o desenvolvimento do modelo de análise biofísica para obtenção da Densidade de Copado Florestal (FCD) usando a análise de imagem do LANDSAT TM. As componentes do modelo FCD eram: vegetação, solo nu, temperatura e sombra. Este trabalho foi implementado pelo projecto de investigação PD32/93Rev2(F) da Organização Internacional para a Madeira Tropical (ITTO).

Key words: Canopy density mapping, FCD mapper, Landsat, remote sensing, semi-expert system.

Introduction

As generally applied in forestry, conventional RS methodology is based on qualitative analysis of information derived from "training areas" (*i.e.* ground-truthing). This has certain disadvantages in terms of the time and cost required for training area establishment, and the accuracy of the results obtained. In response to these problems, new methodology was developed during ITTO Project PD 32/93 Rev. 2 (F), "Rehabilitation of Logged-over Forests in Asia-Pacific Region, Sub-project III" (JOFCA 1991, 1993). In this new methodology, forest status is assessed on the basis of canopy density (JOFCA 1995, 1996). The methodology is presently identified as the Forest Canopy Density

Address for Correspondence: A. Rikimaru, Department of Environmental Engineering, Nagaoka University of Technology, Kamitomioka-Cho, Nagaoka, Niigata-Pref, Japan Zip 940-2188, Tel/Fax: +81-258-47-9668; Email: rikimaru@nagaokaut.ac.jp.

Mapping (FCD) Model. Unlike the conventional qualitative method, the FCD Model indicates the growth phenomena of forests, which is quantitative analysis. (Figs. 1 & 2) The degree of forest density is expressed in percentages: i.e. 10%; 20%; 30%; 40% and so on. FCD data indicates

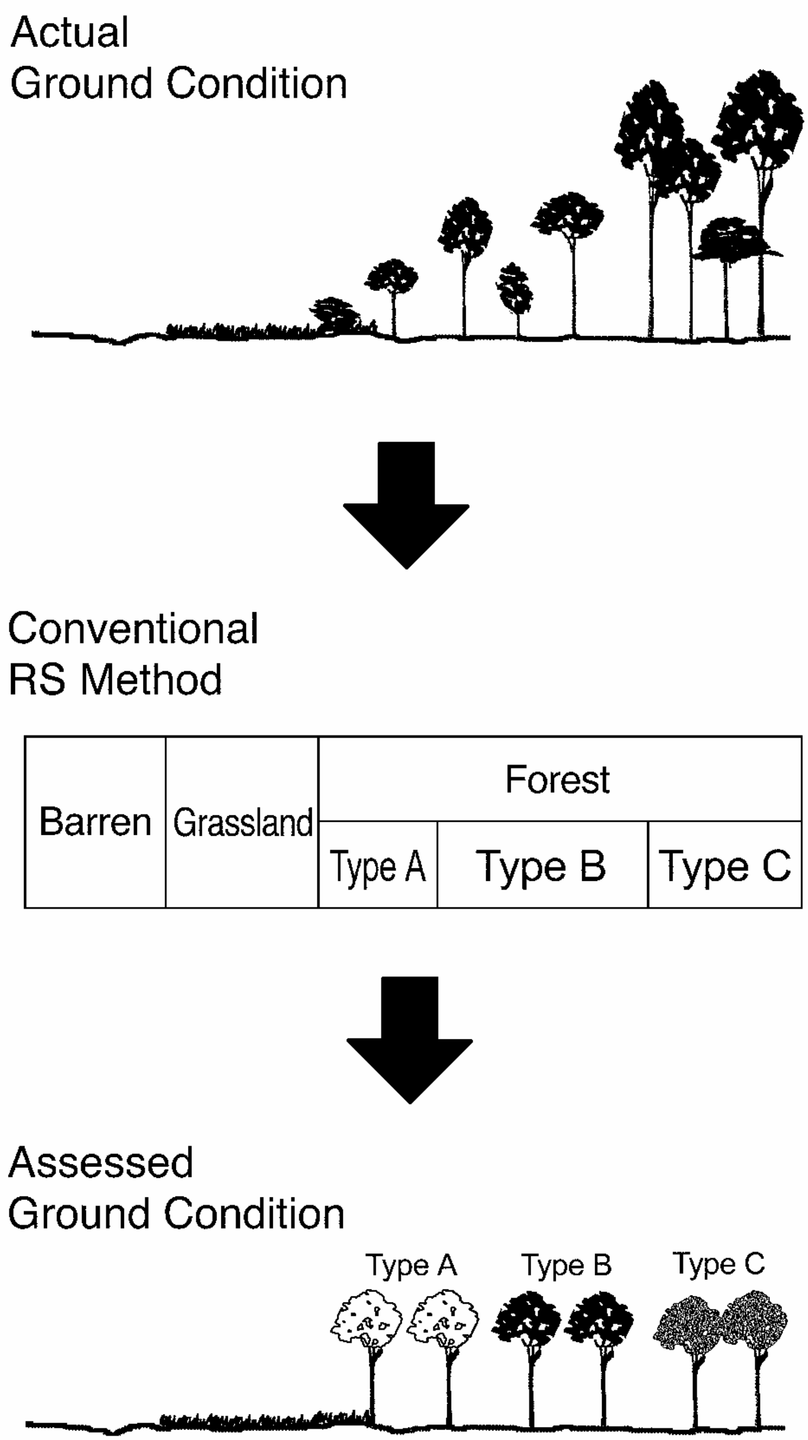

Fig. 1. Analysis by conventional remote sensing method.

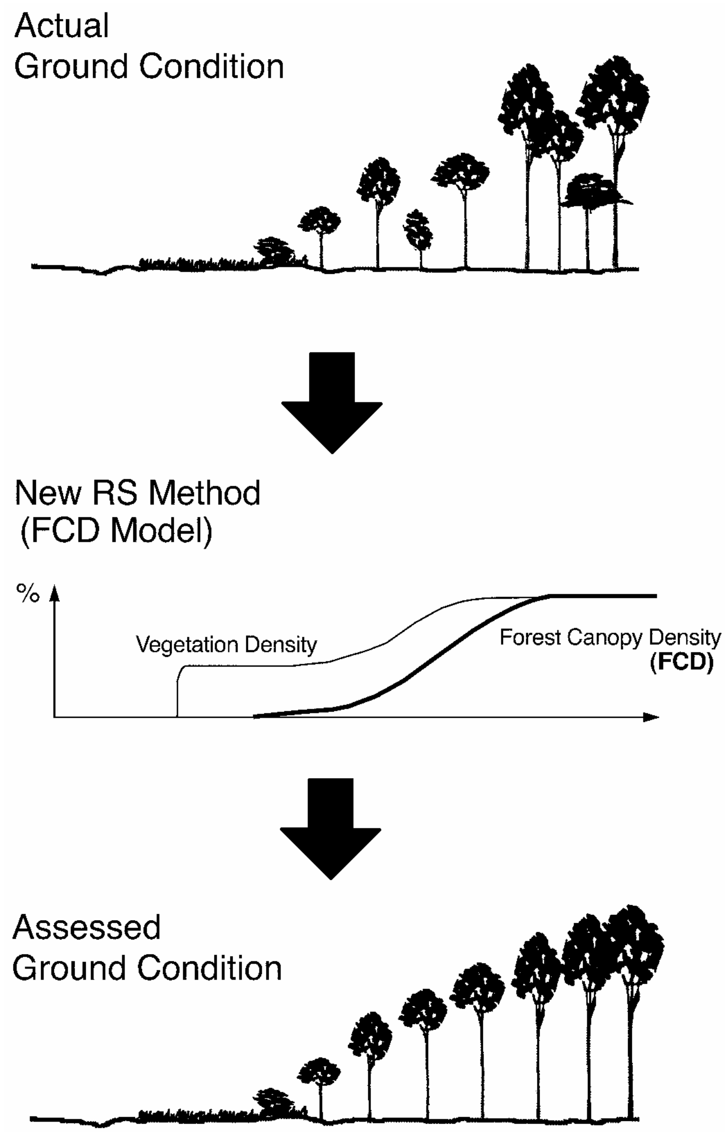

Fig. 2. Analysis by FCD mapping model.

the intensity of rehabilitation treatment that may be required. The method also makes it possible to monitor transformation of forest conditions over time including degradation. Additionally, it can assess the progress of reforestation activities.

The Forest Canopy Density (FCD) Mapping and monitoring Model utilizes forest canopy density as an essential parameter for characterization of forest conditions. FCD data indicates the degree of degradation, thereby also indicating the intensity of rehabilitation treatment that may be required.

The remote sensing data used in FCD model is LANDSAT TM data. The FCD model comprises bio-physical phenomenon modeling and analysis utilizing data derived from four indices: Advanced Vegetation Index (AVI), Bare Soil Index (BI), Shadow Index or Scaled Shadow Index (SI, SSI) and Thermal Index (TI). It determines FCD by modeling operation and obtaining from these indices.

The canopy density is calculated in percentage for each pixel. The FCD model requires less information of ground truth. Just for accuracy check and so on. FCD model is based on the growth phenomenon of forests. Consequently, it also becomes possible to monitor transformation of forest conditions over time such as the progress of forestry activities The application test were implemented in these area. The evergreen forests are in the islands of Luzon (Philippines) and Sumatra (Indonesia); and for monsoon (subtropical deciduous) forests in Ching-Mai (Thailand) and Terai (Nepal).

Characteristics of four indices

The indices have some characteristics as below. The Forest Canopy Density Model combines data from the four indices. Fig. 3 illustrates the relationship between forest conditions and the four indices (VI, BI, SI and TI). Vegetation index response to all of vegetation items such as the forest and the grassland. Advanced vegetation index AVI reacts sensitively for the vegetation quantity compared with NDVI. Shadow index increases as the forest density increases. Thermal index increases as the vegetation quantity increases.Black colored soil area shows a high temperature. Bare soil index increases as the bare soil exposure degrees of ground increase. These index values are calculated for every pixel. Fig.3 shows the characteristics of four indices compared with forest condition.

Note that as the FCD value increases there is a corresponding increase in the SI value. In other words where there is more tree vegetation there is more shadow. Concurrently, if there is less bare soil (i.e. a lower BI value) there will be a corresponding decrease in the TI value. It should be noted that VI is "saturated" earlier than SI. This simply means that the maximum VI values that can be recorded appear earlier in the analysis.

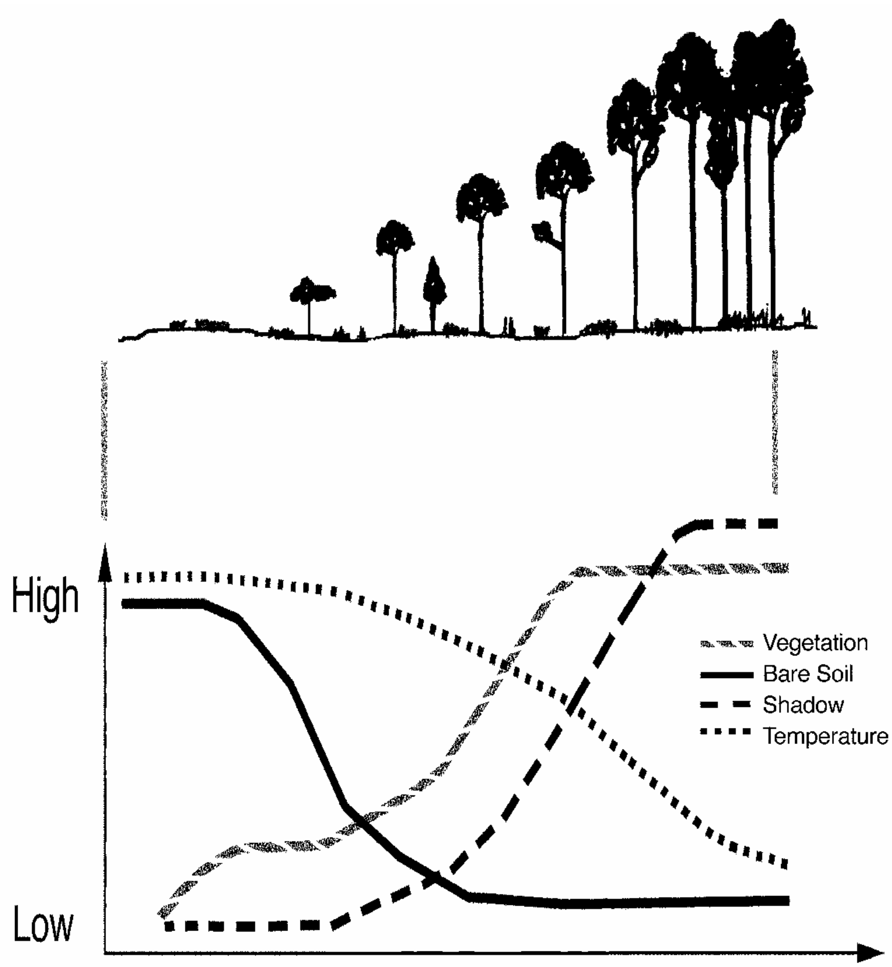

Fig. 3. The characteristics of four indices for forest condition.

This happens because the VI captures data from the total bio-mass, regardless of the density of the trees or forest. On the other hand, the SI values are primarily dependent on the amount of tall vegetation such as trees which cast a significant shadow. The comparative characters of the four indices are shown in Table 1.

Table 1. Combination characteristics between four indices.

	Hi- FCD	Low-FCD	Grass land	Bare Land
AVI	Hi	Mid	Hi	Low
BI	Low	Low	Low	Hi
SI	Hi	Mid	Low	Low
TI	Low	Mid	Mid	Hi

Obtaining of four indices

Advanced vegetation index (AVI)

When assessing the vegetation status of forests, the new methods first examine the characteristics of chlorophyll-a using a new Advanced Vegetation Index (AVI) that is calculated with the following formulae:

B1～B7: TMBand 1～7 data

B43=B4-B3 after normalization of the data range.

B43 < 0 AVI= 0 (Case a)

B43 > 0 AVI= $((B4 +1) \times (256\text{-}B3) \times B43)^{1/3}$ (Case b)

Bare soil index (BI)

The value of the vegetation index is not so reliable in situations where the vegetation covers less than half of the area. For more reliable estimation of the vegetation status, the new methods include a bare soil index (BI) which is formulated with medium infrared information. The underlying logic of this approach is based on the high reciprocity between bare soil status and vegetation status. By combining both vegetation and bare soil indices in the analysis, one may assess the status of forestlands on a continuum ranging from high vegetation conditions to exposed soil conditions.

$$BI = \frac{(B5 + B3) - (B4 + B1)}{(B5 + B3) + (B4 + B1)} \times 100 + 100$$

$(0 < BI < 200)$

The range of BI is converted within 8 bits range

Shadow index (SI)

One unique characteristic of a forest is its three dimensional structure. To extract information on the forest structure from RS data, the new methods examine the characteristics of shadow by utilizing (a) spectral information on the forest shadow itself and (b) thermal information on the forest influenced by shadow. The shadow index is formulated through extraction of the low radiance of visible bands.

$$SI = ((256\text{-}B1) \times (256\text{-}B2) \times (256\text{-}B3))^{1/3}$$

Thermal index (TI)

Two factors account for the relatively cool temperature inside a forest. One is the shielding effect of the forest canopy, which blocks and absorbs energy from the sun. The other is evaporation from the leaf surface, which mitigates warming. Formulation of the thermal index is based on this phenomenon. The source of thermal information is the thermal infrared band of TM data.

The procedure of FCD model

The flowchart of the procedures for FCD mapping model is illustrated in Fig. 4. Image processed results corresponding to the flow chart shows in Fig. 5.

Noise reduction, clouds, cloud shadow and water area

Clouds have a higher irradiant value than ground data. Moreover, the amount of irradiant varies depending on whether the clouds are white, gray, black or combinations of different shades. These factors adversely influence statistical treatment and analysis of imagery data. Furthermore, cloud shadow can be confused with shadow cast by adjacent mountains, These problems can be minimized by creating a cloud shadow mask, using a histogram based on data derived from TM band 1, 2, and 3. Thereafter, a shadow mask of the mountain shadow area is formed at the ground level. This is done through parallel transformation of the mask of the cloud area. Water bodies create similar problems. Since water absorbs near infrared, water bodies should (and can) also be masked using a histogram of TM Band 4.

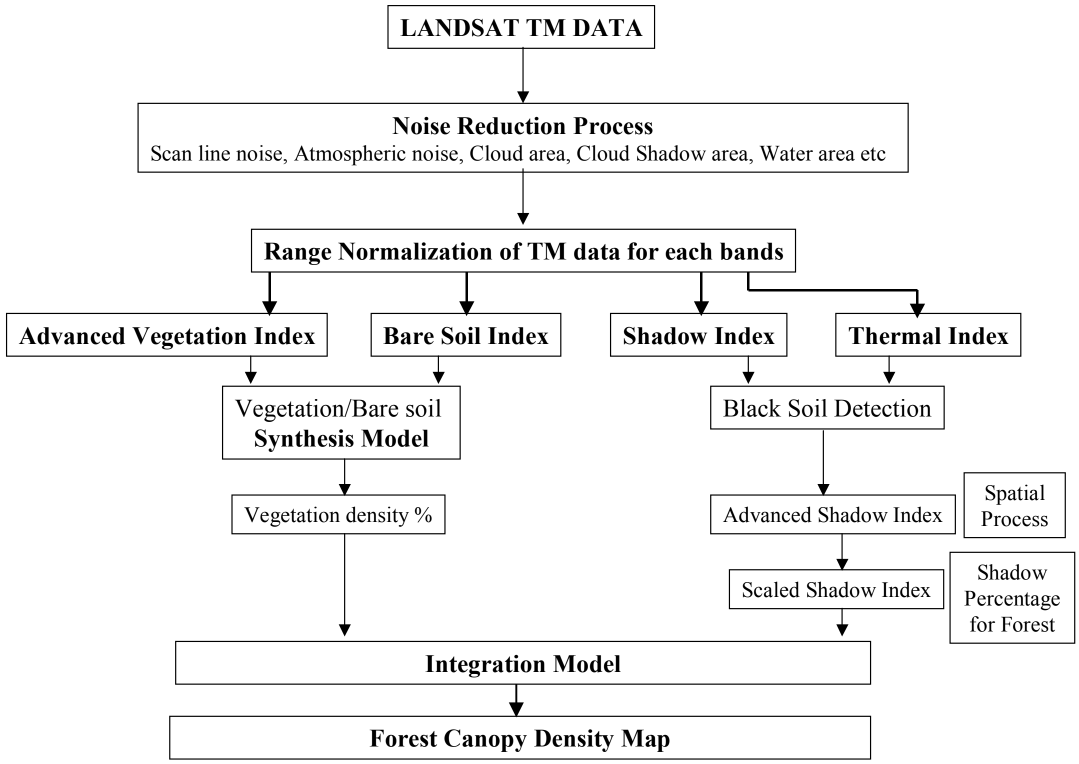

Fig. 4. Flow chart of FCD mapping model

Vegetation density (VD)

It is the procedure to synthesize VI and BI. Processing method is using principal component analysis. Because essentially, VI and BI have high correlation of negative. After that, set the scaling of zero percent point and a hundred percent point. Details are given in Rikimaru (1996).

Black soil detection

SI data is extracted from the low irradiant area of each visible band. Where the soil is black or appears to be black due to recent slash-and-burn, low irradiant data may confuse shadow phenomenon with black soil conditions. This is because black soil usually has high temperature due to its high absorption rate of sun energy. But shadows lead to a decrease in soil temperature. By overlaying TI data and SI data this confusion can be avoided. Overlays are also useful when evaluating the relative irradiance of different parcels of land characterized by various shades of black soil.

Advanced shadow index (ASI)

When the forest canopy is very dense, satellite data is not always be able to indicate the relative intensity of the shadow. Consequently, crown density might be underestimated. To deal with this problem, the new methods include those described below for determining the spatial distribution of shadow information. Details are given in Rikimaru (1996).

Scaled shadow index (SSI)

The shadow index (SI) is a relative value. Its normalized value can be utilized for calculation with other parameters. The SSI was developed in order to integrate VI values and SI values. In areas where the SSI value is zero, this corresponds with forests that have the lowest shadow value (i.e. 0%). In areas where the SSI value is 100, this corresponds with forests that have the highest possible shadow value (i.e. 100%). SSI is obtained by linear transformation of SI.

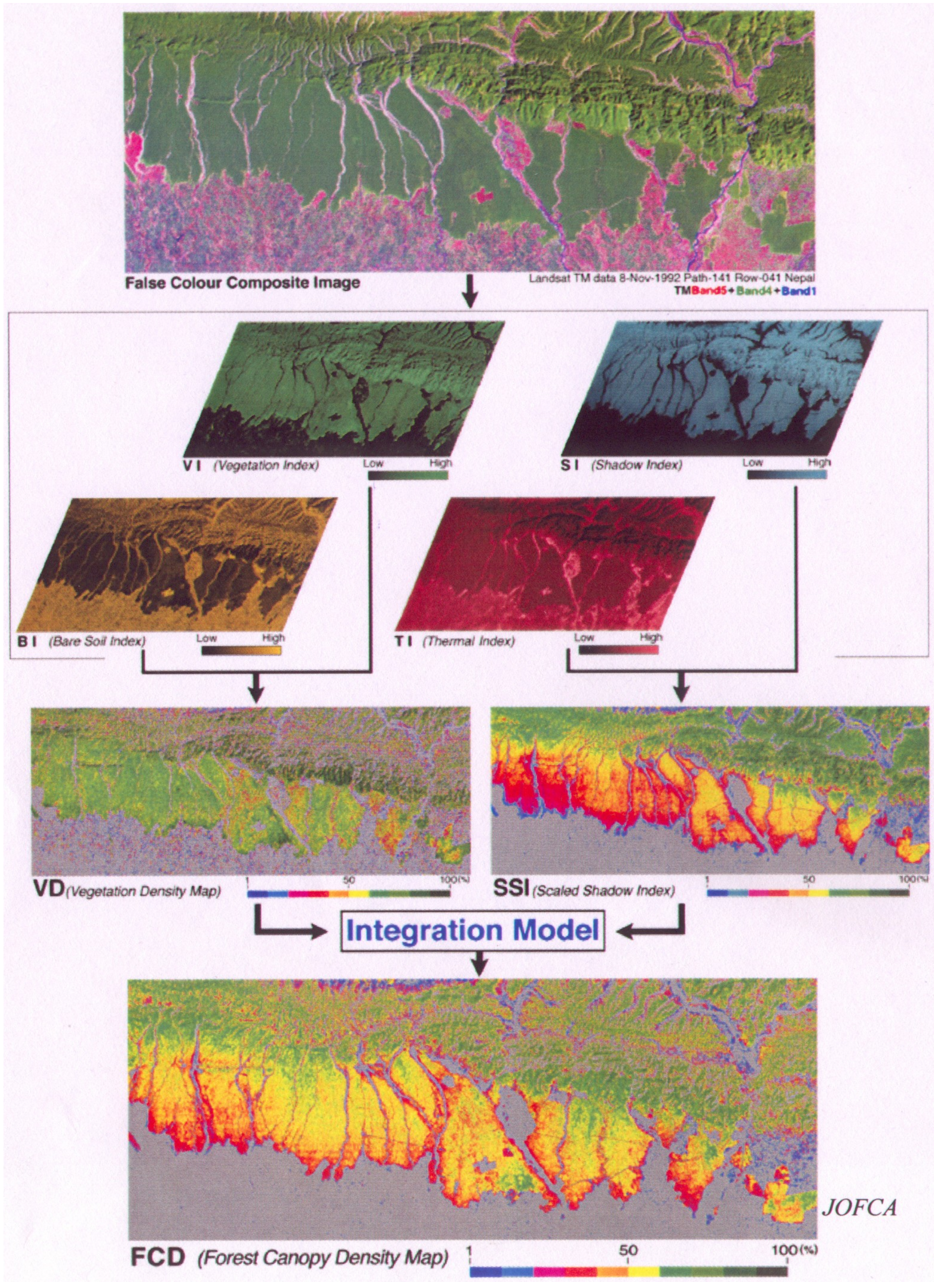

Fig. 5. Procedure for FCD mapping model.

With development of the SSI one can now clearly differentiate between vegetation in the canopy and vegetation on the ground. This constitutes one of the major advantages of the new methods. It significantly improves the capability to provide more accurate results from data analysis than was possible in the past.

Integration process to achieve FCD model

Integration of VD and SSI means transformation for forest canopy density value. Both parameter has no dimension and has percentage scale unit of density. It is possible to synthesize both indices safely by means of corresponding scale and units of each.

$$FCD = (VD \times SSI+1)^{1/2} - 1$$

Development of the semi-expert system for the FCD mapping model

At present, RS assessment of forest conditions is a task reserved for personnel who have received special training in remote sensing analysis. Furthermore, the images derived from conventional RS analysis have rarely been simple enough to be readily understood by practicing foresters, planning officers and other concerned individuals.

As a follow-on phase of the previous project, the Japan Overseas Forestry Consultants Association (JOFCA) has been implementing ITTO Project, PD 13/97 Rev. 1 (F), and developed the Semi-expert System to make the new methodology more available to persons who are not RS experts. The system is now available on CD-ROM format.

Principal features of the semi-expert system

The Semi-expert System, or named the FCD-Mapper, is a computer software package compatible with Windows-type personal computers. The FCD Mapper contains the algorithms and other formulas utilized to compute values of the several indices contained in the FCD Model for the analysis of satellite imagery data. The system will upgrade the planning capacity of decision-makers in forest management and increase the information collecting capability of foresters, while reducing costs and saving time. Use of the Semi-expert system will facilitate production of "user-friendly" satellite imagery that accurately portrays forest conditions. Access to such imagery will increase the capability of foresters to provide accurate, unbiased data on the status of forests in a format that is easy to understand.

The concept of FCD-mapper

The concept of FCD-Mapper is the useful system for the forester, and not for the remote sensing engineer. The Practical Processing System for the Forester which implements FCD Mapping Process as if the Remote Sensing Professional would operate. FCD-Mapper has the knowledge of remote sensing specialist for analysis of FCD model.

Specifications of the semi-expert system

Data Source	:	Landsat TM data
Optional Mode	:	IRS-1C; 1D; SPOT-4, etc.
Output Results	:	FCD Map (Color Hard Copy); Area Counting Statistics of FCD Stratification (Comments of FCD Condition)
Export File Format	:	BMP
Computer	:	DOSV/Windows 95, 98 or NT type
CPU	:	Pentium 133 MHz or above
Memory	:	32MB or above (64MB recommended)
Hard Disk Capacity	:	Free-space 500 MB or above (for processing of TM full scene, 1GB or above recommended)
Image Display	:	1024×768 pixels, 64k color variation or above
Printer	:	Windows compatible type (color/ monochrome)
FCD Model	:	Interactive Operation with Assistance of Semi-expert System
Sub-memory Media	:	MO-Disk 230MB (640MB recommended)

Table 2. Comparison of the semi-expert system with conventional methods.

	Semi-expert System	Work Station	Personal Computer
Processing Time	0.5 days	4~5 days	5~6 days
Operator	Forester	Remote-sensing Expert	Remote-sensing Expert
Time Required for Training	2~3 days	2~3 months	2~3 months
Hardware Cost	$ 3,000~4,000 US	$ 20,000 US~	$ 3,000~4,000 US
Software Cost	Provided for the ITTO Members	$ 10,000 US~	$ 4,000 US~

The core-members of the development team

The FCD Model and FCD Mapper were developed by Dr. Atsushi Rikimaru. This work was carried out in collaboration with Dr. P. S. Roy, Dean of the Indian Institute of Remote Sensing, Dr. Surachai Ratanasermpong, Senior Research Officer of the National Research Council of Thailand, Mr. Ruandha Agung Sugardiman of the Directorate General of Forest and Estate Crops Inventory and Land Use of Indonesia, and Mr. Virgilio Basa and Ms. Alma Arquero of the National Mapping and Resource Information Authority of the Philippines. Computer software programming of the FCD Mapper has been accomplished by Mr. Minoru Nakajima. Continuing improvement in the FCD Model and the Semi-expert System is a collaborative task of all those just mentioned.

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